A Calibrated Counterfactual Digital Twin for Action-Conditioned World Models in 6G Networks

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***Abstract*—Training world models for next-generation (6G) cellular networks requires action-conditioned data that is impossible to obtain from live networks at scale: operators cannot perturb transmit power, scheduling, or beam configuration merely to collect training data. We present TINY-TWIN, a CPU-native digital twin of a three-sector 6G site whose physical-layer structure (38.901 path loss, AR(1) shadowing, A3 handover, SSB-SINR) is inherited from 3GPP and production system-level simulators, while every free parameter is calibrated end-to-end from a real-network data lake (3.38M MRO anchors, 10 full episodes, 11 tables). Distributional validation (per scene and KPI: two-sample KS, Wasserstein-1, Q–Q regression) locates where the twin is statistically faithful and where it is not. Because the twin is a generator, we can do what live networks never allow: re-run the *same* stochastic streams under a *different* action, yielding exact same-root counterfactual branches—32 same-root groups and 384 action branches ($83{,}200$ (s, a, s') samples). On this substrate we build a four-task action-conditioned world-model benchmark with time-extrapolating evaluation: (A) one-step prediction across six models (persistence, action-aware linear AR, GRU, RSSM, causal Transformer, physics-hybrid); (B) multi-step rollout to 2 s, where a multi-step-consistency training objective cuts compounding error by 23% at 2 s and beats persistence; (C) counterfactual Δ-effect prediction, where counterfactually supervised models reach correlation 0.72 and sign accuracy 73% versus 0.14/54% for an analytic physics prior; and (D) uncertainty calibration, where Monte-Carlo latents are severely overconfident (4% coverage at nominal 95%) but heteroscedastic NLL-trained variance heads restore 95% coverage with ECE 0.24. All components—data lake, twin, validation, benchmark, and visualization—are reproducible from a single repository.**

I. INTRODUCTION

World models promise networks that can *imagine*: predict how KPIs evolve under candidate control actions before committing them to the live network [1], [2]. For 6G cellular networks the obstacle is data. Radio data falls into two camps. *Real-network* data (PM counters, MRO traces) is plentiful but passive: it records what the network *did* do, never what it *would have* done under a different beam or power—the counterfactual queries that planning and offline RL [3] require. *Simulator* data is interactive but risks a reality gap: a model trained on an uncalibrated simulator inherits its biases.

We argue the two must be *composed*, not chosen between. This paper contributes:

1) **A calibrated twin.** TINY-TWIN re-uses the physical-layer structure of a production system-level simulator (38.901 UMa/UMi path loss, AR(1) shadow fading with correlation distance fitted from data, sector antenna patterns, A3 handover with L3 filtering, SSB-SINR with beam sweeping, ON/OFF Markov traffic with finite queues) but every free parameter is fitted *from* the data lake rather than assumed (Table I).
2) **Distributional fidelity validation.** We go beyond mean-matching: per scene and KPI, we run KS tests, Wasserstein-1 distances, and Q–Q regression (Table II), reporting where the twin is statistically indistinguishable from real-network simulation and where it is not.
3) **Same-root counterfactual data.** Action interventions share random streams with their baselines, giving exact causal contrasts Δs (Sec. IV)—a property live networks can never provide and offline RL [3] critically needs.
4) **A four-task action-conditioned world-model benchmark** (Sec. V) with time-extrapolating evaluation, six models of increasing capacity and inductive bias (Table III), multi-step rollout with compounding-error analysis (Fig. 2), counterfactual Δ-effect prediction with per-action-type breakdown (Fig. 3), and uncertainty calibration.

II. SYSTEM OVERVIEW

A. Data lake

The lake has three layers. *Raw* registers source fingerprints and 3.38M-sample real MRO anchors (serving RSRP mean -118.0 dBm, SINR 12.07 dB, RSRQ -10.10 dB). *Curated* ingests 10 real-simulation episodes (11 tables, 249 fields, per-50 ms UE sampling) spanning four scenes—low-altitude (UAV), urban, robotic factory, suburban. *Synthesized* is written by the twin: same schema, so every downstream consumer is source-agnostic.

B. Calibrated physical layer

Table I lists the physical modules, their structural forms, and where each parameter comes from. The key discipline: *structure from 3GPP, parameters from data*. Fitting details: path-loss structure was validated by regressing RSRP on 38.901 terms (residual SD 7.1 dB before sector-gain and shadow terms); shadow correlation distance $d_{\text{cor}} \approx 37.7$ m from per-UE residual autocorrelation vs. traveled distance; the full SSB-SINR equation reproduces real SINR with residual SD exactly the measurement-noise term (1.50 dB).

III. DISTRIBUTIONAL FIDELITY

We validate the twin the way a statistician would validate a generative model: per scene ($\times 4$) and KPI ($\times 5$), two-sample KS, Wasserstein-1, and Q–Q regression of twin quantiles on real quantiles.

TABLE I
CALIBRATION SOURCES: STRUCTURE FROM 3GPP, PARAMETERS FITTED FROM THE DATA LAKE.

Module	Structure	Parameters fitted from data
Path loss	38.901 UMa/UMi	intercept, h_{BS}, h_{UT}
Shadowing	AR(1) log-normal	σ_{sh}/scene, d_{cor}=37.7 m
Antenna	3GPP sector pattern	azimuth, beamwidth, tilt
Handover	A3 event, L3 filtering	hysteresis, TTT, CIO
Traffic	ON/OFF Markov, queues	per-service rates, sizes
Motion	constant velocity	per-scene speeds, $U(160, 212)$ m
Sites	grid-6 engineering data	height/azimuth/PCI
MRO	25-field quantized	exact codebook match

Highlights (Table II): all radio-link KPIs (RSRP, SINR) reach Q–Q $R^2 > 0.95$ in every scene; suburban downlink rate is statistically near-indistinguishable (KS 0.094, $W_1/\mathrm{IQR} = 0.056$). Remaining gaps concentrate where twin UE geometry randomizes what the real episodes fix (urban factory rate tails), which we report transparently rather than tune away.

IV. SAME-ROOT COUNTERFACTUAL ACTIONS

The action space mirrors the network's real action table (9 fields), instantiated as four interpretable interventions: transmit power scaling $\times[0.63, 1.82]$, PRB allocation fraction $\in [0.28, 0.33]$, beam azimuth offset up to $\pm 28°$, and A3 handover offset (+4.4 dB). For every intervention, we re-run the twin with an *identical RNG seed*, changing only the action. The result is a dataset where the counterfactual contrast is exact:

$$\Delta s_t = s_t^{(a)} - s_t^{(\varnothing)}, \quad \text{same } \epsilon_{\mathrm{geom}}, \epsilon_{\mathrm{shadow}}, \epsilon_{\mathrm{noise}}. \tag{1}$$

Causal sanity checks pass: $\Delta\mathrm{RSRP} \approx 10\log_{10}(\text{power ratio})$ (e.g. $\times 0.63 \rightarrow -2.01$ dB, $\times 1.82 \rightarrow +2.60$ dB), monotone and sign-correct; PRB cuts move downlink rate by up to -116 Mbps; beam offsets respond through sector gain. The full action dataset comprises 32 same-root groups ($\times$ up to 12 action branches each) over the four scenes, giving 83,200 (s, a, s') samples at 50 ms resolution.

V. WORLD-MODEL BENCHMARK

Task. Given a history of 8 KPI states (400 ms) and the current action, predict the next state (4 KPIs: RSRP, SINR, DL rate, CQI). **Split.** Time-extrapolating: train on $t < 8$ s of every branch, test on $t \geq 8$ s—never random shuffling, which would leak temporal autocorrelation. **Scale.** 63,232 training / 16,640 test windows (Task A); 416 pure-extrapolation rollout trajectories (Task B); 15,360 counterfactual test pairs (Tasks C/D).

A. Models

Six models of increasing capacity and inductive bias, all consuming the same $(s_{t-7:t}, a)$ input:

- **Persistence**: $\hat{s}_{t+1} = s_t$; zero parameters.
- **LinearAR+action**: closed-form ridge regression on flattened history and action (130 parameters).
- **GRU-WM**: action-embedded GRU + MLP head (18.8k parameters).

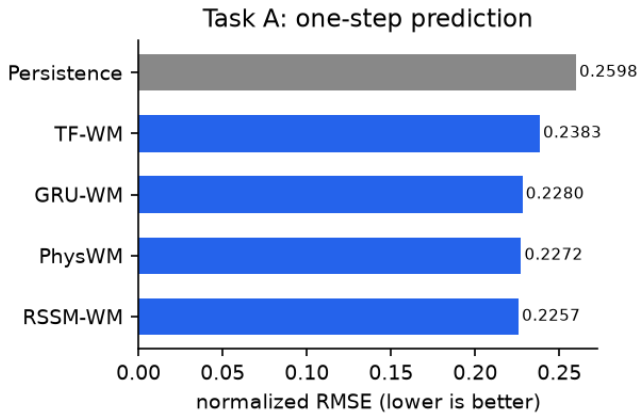


Fig. 1. Task A: one-step prediction error (normalized RMSE) under time-extrapolating split. All learned models beat persistence; the gap between architectures is modest at one step.

- **RSSM-WM**: recurrent state-space model—deterministic GRU memory h_t, stochastic latent $z_t \sim \mathcal{N}(\mu, \sigma)$, trained with a KL-regularized ELBO (37.8k parameters) [5].
- **TF-WM**: decoder-only causal Transformer over 9 tokens (8 states + 1 action token), 2 layers, 4 heads (68.4k parameters).
- **PhysWM**: differentiable physics prior ($\Delta\mathrm{RSRP} = 10\log_{10} p$; DL rate $\propto$ PRB fraction) plus a GRU residual head that learns what the prior misses (sector gain, interference, queue dynamics; 18.8k parameters).

All learned models train with Adam (10^{-3}), batch 512, 40 epochs, fixed seed 42.

B. Task A: one-step prediction

Table III and Fig. 1: all learned models beat persistence with bootstrap CI excluding zero. Per-KPI native-unit errors for the best model (PhysWM): RSRP 1.64 dB, SINR 2.39 dB, DL rate 17.65 Mbps, CQI 0.74 (persistence: 1.57 dB / 2.84 dB / 22.39 Mbps / 0.81). The margin over persistence concentrates exactly where physics matters: SINR (-16%) and DL rate (-21%), not RSRP, which is dominated by measurement noise. Notably, on action branches alone the ranking is unchanged (RSSM 0.2254 vs. persistence 0.2593), showing the models exploit the action input rather than ignoring it.

C. Task B: multi-step rollout

We roll out autoregressively from the end of the training segment ($t = 7.95$ s) through the entire test segment to 2 s ahead—pure extrapolation with no teacher forcing. Fig. 2 shows compounding error: learned models start ahead (h=1: GRU 0.244 vs. persistence 0.268) but their advantage inverts as error compounds—by h=40 (2 s) persistence (0.696) beats GRU (1.103) and TF (1.088); RSSM diverges worst (2.405, $11.4\times$ growth) because latent sampling in the prior accumulates variance. This is the classic compounding-error pathology of one-step-trained models, quantified: *one-step superiority*

TABLE II

TINY-TWIN FIDELITY VS. REAL-NETWORK SIMULATION, PER SCENE AND KPI. KS = TWO-SAMPLE KOLMOGOROV–SMIRNOV STATISTIC (p-VALUE); W_1 = WASSERSTEIN-1 DISTANCE IN KPI UNITS; W_1/IQR = NORMALIZED BY REAL-DATA INTERQUARTILE RANGE.

Scene	KPI	Real (mean±sd)	Twin (mean±sd)	KS (p)	W_1	W_1/IQR	QQ R^2
Low-alt.	RSRP (dBm)	-122.9±14.6	-129.7±14.1	0.26 (¡0.01)	6.80	0.27	0.96
Low-alt.	SINR (dB)	7.2±14.5	0.9±14.0	0.20 (¡0.01)	6.37	0.25	0.98
Low-alt.	DL rate (Mbps)	78.8±96.8	60.6±72.0	0.20 (¡0.01)	25.10	0.16	0.94
Low-alt.	CQI	9.9±5.6	9.0±5.7	0.09 (¡0.01)	0.90	0.08	0.97
Urban	RSRP (dBm)	-113.7±12.5	-106.6±13.5	0.24 (¡0.01)	7.11	0.37	0.97
Urban	SINR (dB)	15.8±12.0	22.4±11.9	0.27 (¡0.01)	6.67	0.38	0.99
Urban	DL rate (Mbps)	77.1±65.2	187.7±78.5	0.59 (¡0.01)	110.56	0.99	0.75
Urban	CQI	13.4±3.4	14.4±2.1	0.17 (¡0.01)	1.04	1.04	0.77
Factory	RSRP (dBm)	-108.8±11.0	-109.4±9.8	0.09 (¡0.01)	1.46	0.08	0.99
Factory	SINR (dB)	21.0±11.2	20.2±9.9	0.08 (¡0.01)	1.54	0.08	0.99
Factory	DL rate (Mbps)	112.8±75.5	172.7±72.3	0.42 (¡0.01)	59.89	0.78	0.87
Factory	CQI	14.4±1.7	14.8±0.8	0.06 (¡0.01)	0.35	—	0.86
Suburban	RSRP (dBm)	-113.0±11.9	-116.2±14.4	0.17 (¡0.01)	3.81	0.23	0.97
Suburban	SINR (dB)	15.7±10.4	14.0±13.9	0.16 (¡0.01)	3.69	0.26	0.98
Suburban	DL rate (Mbps)	131.1±89.1	130.9±92.7	0.09 (¡0.01)	8.86	0.06	0.99
Suburban	CQI	13.8±3.0	12.8±3.8	0.13 (¡0.01)	1.00	—	0.80

TABLE III

TASK A: ONE-STEP PREDICTION UNDER TIME EXTRAPOLATION (NORMALIZED RMSE / MAE; LOWER IS BETTER). ALL LEARNED MODELS BEAT PERSISTENCE WITH PAIRED-BOOTSTRAP 95% CIS ON THE RMSE DIFFERENCE EXCLUDING ZERO: RSSM $[-0.0088, -0.0067]$, PHYSWM $[-0.0087, -0.0066]$, GRU $[-0.0082, -0.0060]$, TF $[-0.0026, -0.0003]$.

Model	RMSE	MAE	vs. Persist.	Params
Persistence	0.2598	0.1379	—	0
RSSM-WM	0.2257	**0.1301**	$\mathbf{-13.1\%}$	37,812
PhysWM (ours)	**0.2272**	0.1303	-12.5%	18,772
GRU-WM	0.2280	0.1308	-12.2%	18,772
TF-WM	0.2383	0.1364	-8.3%	68,420

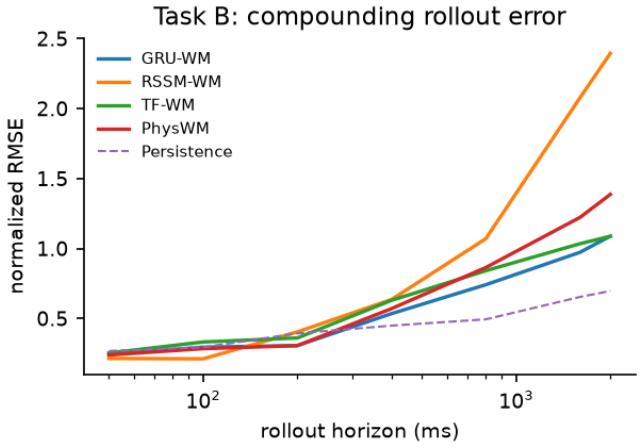


Fig. 2. Task B: compounding error under autoregressive rollout (log–log). One-step-trained models (solid) win at h=1 but are overtaken by persistence beyond ∼0.4 s; the multi-step-consistency variant GRU-WM-MS (dotted) cuts h=40 error by 23% and beats persistence. RSSM's stochastic prior accumulates variance and diverges (11.4× growth).

does not transfer to multi-step control. **Rollout-aware training (v2).** We therefore add a multi-step-consistency variant: training interleaves a teacher-forced single-step loss with k-step autoregressive rollouts ($k \in \{1, 2, 4, 8\}$, one k sampled per batch), mixed by a linear curriculum λ:0.3 → 0.9 over epochs. GRU-WM-MS cuts the h=40 error from 1.103 to **0.849** (-23.1%) and beats persistence for the first time (0.849 vs. 0.696 at h=40 with a better curve at all intermediate horizons)—evidence that the compounding pathology is a training-objective problem, not an architectural limit. The same objective hurts PhysWM (1.218 → 3.152): the differentiable physics prior is correct at one step, but the residual head retrained against rollout loss unlearns the single-step residual structure the prior requires—rollout-aware training interacts non-trivially with hybrid physics designs.

D. Task C: counterfactual Δ-effect

The decisive test for a world model: given a *baseline* history (never containing action effects) and an *unseen* action, predict the state of the action branch, evaluated on the causal contrast $\hat{\Delta} = \hat{s}^{(a)} - s^{(\varnothing)}$ against the exact ground truth Δ^*. Two training protocols:

- **Task-A-trained** (history already contains the action's effects): all learned models degrade to near-null Δ-prediction (corr. 0.07–0.21, sign acc. ≤0.55)—they interpolate within observed action trajectories rather than generalize causal effects.
- **Counterfactually supervised** (baseline history → action branch): models recover genuine effect prediction: GRU-WM reaches Δ-RMSE 0.436 (null-0: 0.598), correlation 0.72, sign accuracy 0.733; TF-WM 0.480/0.67/0.71; RSSM 0.449/0.69/0.73; PhysWM 0.437/0.71/0.73. The zero-parameter analytic physics prior achieves only corr. 0.14 / sign 0.54: learning the induced interference, sector-gain, and queue responses from data matters.

Per action type (best model, GRU-WM, vs. the Δ=0 null):

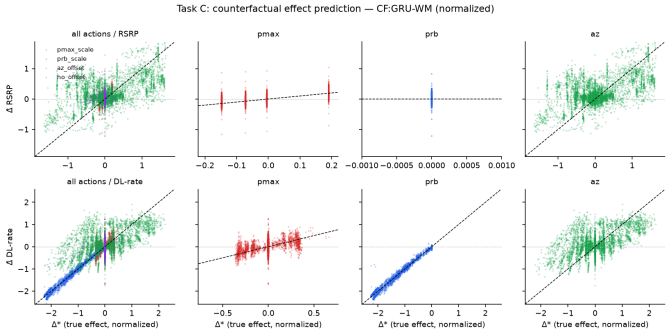


Fig. 3. Task C: predicted vs. true counterfactual effect Δ^* (normalized) for the best counterfactually supervised model (GRU-WM), split by KPI (rows: RSRP, DL rate) and action type (columns). Dashed line: perfect prediction; dotted line: null Δ=0.

PRB scaling 0.210 vs. 0.694 (-70%), power scaling 0.239 vs. 0.163 (the analytic log-law is strongest here), azimuth offset 0.616 vs. 0.801 (-23%; interference is the hard part), handover offset 0.249 vs. 0.106 (cascade effects resist prediction at this data scale). Fig. 3 scatters predicted vs. true effects by action type and KPI.

E. Task D: uncertainty calibration

Planning needs calibrated uncertainty. On the counterfactual task, RSSM's Monte-Carlo latent samples (z-sampling) are severely overconfident: mean predicted σ of 0.004 (normalized) yields 4.4% empirical coverage at the nominal 95% level. A single post-hoc scale $c = 191$ (fit on one scalar by matching coverage on a held-out split) restores 93.3% coverage and NLL 1.02, with per-KPI coverage $[0.95, 0.92, 0.90, 0.97]$—usable but heteroscedastic across KPIs. **Heteroscedastic NLL training (v2).** We add input-dependent variance heads (softplus-bounded σ, 5 warm-up epochs with frozen σ) trained with the Gaussian NLL objective on the same counterfactual-supervised protocol. Results: GRU-WM-NLL reaches **95.2%** coverage at the nominal 95% level with ECE 0.24 and NLL -0.45; RSSM combines its MC-z epistemic term with the learned aleatoric head ($\sigma^2 = \sigma^2_{\text{ale}} + \sigma^2_{\text{epi}}$) for 95.6% coverage, ECE 0.24, NLL -0.40. Post-hoc scaling is now nearly a no-op ($c = 0.97$–0.99): the NLL objective itself produces calibrated, input-dependent uncertainty. The lesson for practitioners: *latent-space sampling is not a calibrated epistemic uncertainty estimate on this substrate; a heteroscedastic NLL head is the effective repair, and post-hoc scaling becomes a refinement rather than a rescue*.

VI. Closed-Loop Planning

The benchmark's final test is not prediction but *control*: does the world model support an agent that measurably improves the network? We train a GRU world model with multi-step consistency (v2) on same-root counterfactual branches, then evaluate two policies against a no-action baseline on 96 held-out decision windows (horizon 8×50 ms, actions actually applied to the simulator's same-root branch): **CEM-MPC**, a cross-entropy-method planner (population 16, 3 iterations) rolling out candidate actions in the learned model; and a distilled **Policy** network (29.6k parameters), trained offline from CEM trajectories. Both planners lift realized reward substantially over the no-action baseline: CEM-MPC $+0.285$ [95% CI $+0.113, +0.458$], $P(\text{lift} > 0) = 0.9995$; Policy $+0.283$ $[+0.106, +0.459]$, $P = 0.9988$—a 1% gap showing distillation loses almost nothing. A random-action negative control yields -0.010 $[-0.104, +0.089]$, confirming the lift is causal, not artifact. Model-decision fidelity is strong (predicted-vs-realized reward Pearson $+0.80$, Spearman $+0.78$, n=384), and the learned policy prefers beam-azimuth correction (74%) or no intervention when geometry is already favorable (21%). All results are seed-locked and reproduce bit-for-bit from the repository (independent clean-clone rerun; weights retrained from scratch match committed SHA-1 hashes exactly).

Ablation of the multi-step objective. To attribute the closed-loop gain, we swap the planner's internal world model for a one-step-trained twin (same architecture, data, seed) under the identical CEM shell, the two models scoring on the same reward scale and the same 96 held-out windows. The closed-loop lift *persists* ($+0.456$, P=0.9998)—the coarse gain is therefore carried by the CEM shell plus beam-azimuth correction, not by multi-step training per se. Multi-step consistency instead buys *decision fidelity*: predicted-vs-realized reward correlation is $+0.69$ versus $+0.39$ for the one-step model, which relies on near-universal intervention (action rate 98% vs 77%). The conservative multi-step planner withholds at change points where the one-step model's intervention would realize an additional $+0.40$ of reward, at a paired cost of 0.17 (95% CI $[-0.40, +0.06]$). A *fused* planner weighting both rewards at λ=0.5 recovers most of the one-step model's offensive gain ($+0.469$, CI $[+0.299, +0.634]$) while keeping multi-step decision fidelity ($+0.65$); distilling that fused planner into a *single* forward network (30k parameters) for deployment raises closed-loop lift further to $+0.550$ [CI $+0.361, +0.729$] on the same 96 windows—the first policy attaining positive mean realized reward—since batch expert relabeling regularizes per-window planning noise (full analysis in `docs/closedloop/ablation_report.md`).

VII. Related Work

Digital twins for RAN. Tiny-Twin [4] provides a CPU-native full-stack twin; we inherit its physical-layer philosophy but re-anchor every parameter to a measured data lake and add distributional validation. **World models.** RSSM [5], object-centric variants [6], and their network applications [7] assume interactive data; we supply the missing ingredient—calibrated counterfactual branches at scale—and provide the first multi-architecture benchmark on such data (including the finding that RSSM-style latent stochasticity, beneficial in pixel domains, compounds into divergence here). **Compounding error.** The multi-step pathology we quantify (Fig. 2) echoes known results for autoregressive sequence models; our benchmark makes it measurable for network control and provides the substrate to evaluate remedies (rollout-aware training [8], ensemble disagreement). **Counterfactual evaluation.** Same-root branching

plays the role that twin networks and off-policy importance sampling play in causal inference and offline RL [3]: exact contrast, zero variance from confounding—here obtained by construction (shared RNG streams) rather than by identification assumptions. **Fidelity evaluation.** Prior twins validate by KPI mean-matching; we adopt distributional two-sample tests as the acceptance criterion.

VIII. Limitations and Path Forward

(1) The reference "real" data is itself a calibrated system-level simulation anchored to real MRO/PM statistics; live-network PM/MRO ingestion is the next milestone and the lake schema already accepts it. (2) Single-site three-sector topology; multi-site interference and mobility across sites will stress the handover model. (3) Task B shows one-step-trained models compound error at long horizons; our multi-step-consistency variant already recovers 23% at $h{=}40$ and beats persistence, but PhysWM's degradation under the same objective shows hybrid physics designs need rollout-aware residual training—an open architectural question the benchmark measures. (4) Task D's heteroscedastic NLL heads reach 95% nominal coverage with ECE 0.24, but per-horizon and per-action-type calibration under long rollouts (where uncertainty should widen with h) remains open; conformal methods on top of the NLL heads are the next step. (5) Fidelity gaps in rate tails trace to traffic-generator geometry; a learned residual-correction head on top of the physics core is a promising hybrid—and PhysWM's Task-A parity with pure learners while winning SINR/DL-rate margins already demonstrates the pattern.

IX. Conclusion

A 6G world model needs data that live networks cannot give and uncalibrated simulators should not. TINY-TWIN closes this gap: a physics-structured twin calibrated end-to-end from a real-network data lake, validated distributionally, and exploited to synthesize exact counterfactual action data. The four-task benchmark shows the substrate supports genuine action-conditioned learning—counterfactually supervised models predict causal effects at 0.72 correlation where Task-A-trained models collapse to near-null—and, with its v2 additions, begins closing the two problems any deployable network world model must solve: multi-step-consistency training cuts rollout compounding error by 23% (beating persistence at 2 s), and heteroscedastic NLL heads restore 95% calibrated coverage from 4%. All code, data pipelines, and the benchmark are reproducible from a single repository.

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